Math 1232 Summer 2024
Mastery Quiz 2
Due Tuesday, July 8

This mastery quiz has three topics. Please do your best, but don't worry if you make a minor error—your goal is to demonstrate your mastery of the underlying material.

Feel free to consult your notes, but please don't discuss the actual quiz questions with other students in the course.

Remember that you are trying to demonstrate that you understand the concepts involved. For all these problems, justify your answers and explain how you reached them. Do not just write "yes" or "no" or give a single number.

Please turn this quiz in class on Tuesday. You may print this document out and write on it, or you may submit your work on separate paper; in either case make sure your name is clearly on it. If you absolutely cannot turn it in in person, you can submit it electronically but this should be a last resort.

Topics on this quiz:

- Major Topic 1: Transcendental Functions
- Secondary Topic 1: Invertible Functions
- Secondary Topic 2: L'Hôpital's Rule

Name:

Name: ______________________________

Major Topic 1: Transcendental Functions

(a) Compute $\dfrac{d}{dx}x^{\ln(x)}$.

(b) Compute $\dfrac{d}{dx}\dfrac{1}{\arcsin(x^2)}$.

(c) Compute $\displaystyle\int_0^2 \frac{e^{2x}}{e^{4x}+1}\,dx$.

Name:

Secondary Topic 1: Invertible Functions

(a) Find a formula for the inverse of $g(x) = (x-1)^3 + 3$.

(b) Compute $\log_3(90) + \log_3(3/2) - \log_3(5)$

(c) Let $h(x) = x^7 + 3x^3 + 1$. Compute $(h^{-1})'(5)$.

Name:

Secondary Topic 2: L'Hôpital's Rule

(a) Compute $\displaystyle\lim_{x\to 2} \frac{e^{(x^2-4)} - x + 1}{x - 2}$.

(b) Compute $\displaystyle\lim_{x\to 0} \frac{x^3 - x^2}{x + \sin(x)}$.

(c) Compute $\displaystyle\lim_{x\to\infty} x^{\frac{3}{2+\ln(x)}}$.